SmartClass Vision

Enterprise Automated Classroom Biometrics & Security Architecture

Next-Gen Edge AI Attendance System

Powered by OpenCV YuNet • InsightFace ArcFace • Cryptographic Auditing

Domain: Computer Vision / Edge AI	**Key Models:** YuNet (5-Pt) + ArcFace (512-D)
Accuracy: > 99.8% LFW Benchmark	**Security:** Anti-Spoofing + HMAC-SHA256 Signatures

1. Problem Statement & The Smart Solution

Why Traditional Attendance Fails & How AI Reinvents It

Traditional Methods (The Flaws)	SmartClass Vision (The AI Solution)
• **Time Waste:** Manual roll call consumes 10-15 minutes of every lecture. • **Proxy Attendance:** Friends sign sheets or swipe RFID cards for absentees. • **Single-Photo Vulnerability:** Mobile scans easily tricked by printed photos. • **Unenforced Section Control:** Students attend wrong lecture sections undetected.	• **Zero Instructional Delay:** Automated 10-shot batch scan takes seconds. • **Proxy-Proof AI Voting:** Multi-shot temporal voting requires 5/10 match consensus. • **Anti-Spoofing Defense:** Moiré & frequency analysis rejects phone/paper attacks. • **Strict Section Isolation:** Automatically rejects non-enrolled students.

2. End-to-End AI Pipeline Architecture

From Raw Multi-Student Classroom Capture to Cryptographic Attendance Record

Stage 1: Ingestion	Stage 2: Detection	Stage 3: Quality Gate	Stage 4: ArcFace Match	Stage 5: Consensus
10-Shot Burst Synchronized live classroom camera feed captures 10 rapid frames with audio/visual feedback.	**OpenCV YuNet** Locates all faces + extracts 5 landmarks per face. Applies **affine rotation alignment**.	**Anti-Spoof Filter** Verifies Laplacian sharpness (≥40), pose yaw (≤1.4), and frequency Moiré screen detection.	**512-D Embedding** Vectorized inference against enrolled roster with Cosine ≥ 0.65 and Margin ≥ 0.12.	**Voting & Export** Requires ≥5/10 votes + section roster match. Exports HMAC-SHA256 signed CSV.

3. Why These Specific Models? (Technical Justification)

Deep Dive into Model Selection: OpenCV YuNet vs. ArcFace

Component	Selected Model	Alternative Considered	Why Our Choice is Superior
Face Detection & Landmark Regression	**OpenCV YuNet** (2023 ONNX)	YOLOv8-Face / Haar Cascades	Runs in native C++ DNN with sub-15ms latency on standard CPU. Directly regresses 5 key landmarks (eyes, nose, mouth corners) enabling exact **affine facial alignment**.
Biometric Feature Embedding	**InsightFace ArcFace** (ResNet-50 / 512-D)	FaceNet (Triplet Loss) / dlib (HOG/ResNet)	ArcFace introduces an **Additive Angular Margin** penalty on the hypersphere, forcing intra-class compact clusters and large inter-class margins. Outperforms Euclidean Triplet Loss with **>99.8% LFW accuracy**.

4. Key Parameters & Thresholds (To Remember)

Core Numeric Benchmarks Enforced During Execution

Metric / Parameter	Value	Engineering Rationale
ArcFace Cosine Similarity Threshold	**≥ 0.65**	Ensures zero false positives across multi-student classroom environments.
Top-1 vs. Top-2 Margin Gap	**≥ 0.12**	Ambiguity rejection gate: Prevents confusing siblings or lookalikes.
Multi-Shot Burst Size	**10 Shots**	Captures full classroom view with variable student head positions.
Temporal Consensus Quorum	**≥ 5 / 10 Votes**	Eliminates accidental passers-by outside doors or brief occlusions.
Minimum Sharpness (Laplacian)	**40.0**	Filters out blurry, motion-corrupted face crops before inference.
Biometric Enrollment Profile	**5 Guided Angles**	Captures Center, Left, Right, Up, and Down angles during registration.

5. Security Architecture & Recent Updates

Multi-Tiered Security & Enterprise Features Implemented

1. Section-Specific Validation

- Attendance is strictly bound to Degree, Year, Branch, and Section.
- Non-enrolled students are tagged **REJECTED - WRONG CLASS**.

2. Cryptographic HMAC-SHA256 Signatures

- Every CSV export is sealed with HMAC-SHA256 using faculty credentials.
- Any manual offline tampering permanently breaks verification.

3. Automated Security Event Logging

- All spoof attempts, quality rejections, and attendance events are committed to SQLite.
- Full audit trail for administrators and faculty review.

4. Complete Modernization

- Deprecated FaceNet and YOLO; unified under ArcFace 512-D embeddings.
- High-efficiency OpenCV streaming overlays with real-time feedback.

6. Summary & Viva Takeaways

Key Highlights to State During Your Presentation

1. **Problem Solved:** Replaces 15-minute manual roll calls and proxy-vulnerable cards with automated, proxy-proof, 10-second batch biometric scans.
2. **Winning Model Pair: OpenCV YuNet** for 15ms 5-landmark face alignment + **InsightFace ArcFace** for 512-D angular margin biometric matching.
3. **Robust Defense:** Hardware-independent anti-spoofing (Moiré pattern rejection) + strict Section Validation + HMAC-SHA256 cryptographic signatures.
4. **Ready for Scale:** SQLite backend, vectorized multi-student matrix evaluation, and full audit logging.

Thank You • Questions & Discussion